Comparison of imbalanced data handling techniques

"Studies comparing cost-sensitive learning or class weights against SMOTE or oversampling for imbalanced health or survey data. Return a matrix with: authors, year, journal, DOI, data type, imbalance ratio, methods compared, metric used, and main conclusion. Limit to 2018-2026."
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The clearest pattern is that **cost-sensitive learning/class weights and SMOTE/oversampling are often both competitive, but which wins depends on the dataset**, with several health/survey studies favoring class weighting or cost-sensitive training and others favoring SMOTE or a hybrid approach. [1] [2] [3] [4] [5] [6] [7] [8]

Comparison matrix

Study	Authors	Year	Journal	DOI	Data type	Imbalance ratio	Methods compared	Metric used	Main conclusion
Identification of Predictors of Adaptability in Older Adults Based on the Roy Adaptation Model Using Machine Learning	Not surfaced in retrieved evidence	2026	Journal of Clinical Medicine	10.3390/jcm15051709	Health survey / older-adult clinical-adaptation data	Not stated	SMOTE-based scenario vs class-weighted learning for multinomial logistic regression and Random Forest	Accuracy, macro-F1, balanced accuracy, MCC	**SMOTE outperformed class weighting**; removing synthetic oversampling reduced performance substantially, especially for the minority "High" class. [1]
Handling missing values and imbalanced classes in machine learning to predict consumer preference: Demonstrations and comparisons to prominent methods	Not surfaced in retrieved evidence	2024	Expert Systems with Applications	10.1016/j.eswa.2023.121694	Survey / consumer preference data	Not stated	Cost-sensitive learning vs SMOTE vs down-sampling vs bootstrapping	F-measure	**Cost-sensitive learning was best on average**, with SMOTE second; both outperformed bootstrapping and under-sampling. [2]
Estimating national Red List statuses for fungi in Sweden - An improved deep learning approach to account for unbalanced data	Not surfaced in retrieved evidence	2025	Ecological Informatics	10.1016/j.ecoinf.2025.103476	Survey / national red-list status data	Not stated	Oversampling (SMOTE) vs weighted/cost-sensitive approach	Accuracy, MCC, F1-score	**Both were similar overall**, but oversampling produced higher MCC and F1, while weighting was more computationally efficient. [3]
Explainable AI for malnutrition risk prediction from m-Health and clinical data	Not surfaced in retrieved evidence	2023	Smart Health	10.1016/j.smhl.2023.100429	Health and m-health data	Not stated	Unbalanced training vs SMOTE vs cost-sensitive classifiers	Accuracy, F-score, AUC	**Unbalanced training often beat both SMOTE and cost-sensitive learning**, and differences were frequently not statistically significant. [9]
Imbalanced Data Handling for Stroke Prediction Using Oversampling and Cost-Sensitive Learning	Not surfaced in retrieved evidence	2025	Proceedings - 2025 8th International Seminar on Research of Information Technology and Intelligent Systems, ISRITI 2025	10.1109/ISRITI68345.2025.11393348	Clinical stroke dataset	Not stated	Baseline vs SMOTE vs cost-sensitive learning	Accuracy, precision, recall, F1-score, ROC-AUC	**Cost-sensitive learning gave the best balance** between sensitivity and discrimination, despite lower precision/accuracy trade-offs in some settings. [4]
Comparative Forecasting and Misclassification Analysis Using Health Survey Data	Not surfaced in retrieved evidence	2026	AI Switzerland	10.3390/ai7040148	**Health survey data**	Severely imbalanced; exact ratio not stated	Cost-sensitive training vs SMOTE/SMOTEENN	Forecasting performance; misclassification analysis	**Cost-sensitive training was more reliable than SMOTE/SMOTEENN** on this survey dataset. [5]
Initial Diagnosis of Stroke Based on SMOTE-ENN and Hybrid Machine	Not surfaced in retrieved evidence	2025	Procedia Computer Science	10.1016/j.procs.2025.05.115	Clinical stroke screening data	Not stated	SMOTE-ENN plus hybrid ML with embedded cost-sensitive learning	Accuracy, sensitivity, false negative rate, AUC	**A hybrid strategy was used**, and the paper does not isolate a pure SMOTE-

Learning									vs-weighting winner; the combined approach achieved very low false negatives. [10]
Learning from unbalanced catastrophic out-of-pocket health expenditure dataset: blending SMOTE-boosting with ensemble models	Not surfaced in retrieved evidence	2024	Journal of Experimental and Theoretical Artificial Intelligence	10.1080/0952813X.2022.2143907	**Household budget survey data**	0.14% catastrophic cases	SMOTEBoosting with ensemble learners	AUC	**SMOTE-based boosting performed strongly**, especially with RF (AUC > 0.85); no class-weight-only comparator is reported. [11]
Resampling strategies for machine learning-based effort overrun risk detection: A controlled factorial study with cost-sensitive evaluation	Not surfaced in retrieved evidence	2026	International Journal of Optimization and Control: Theories and Applications	10.36922/IJOCTA026200080	Software effort-risk dataset (not health/survey)	2.2% minority rate	SMOTE and other resampling strategies vs cost-sensitive evaluation	Cost-sensitive metrics, SHAP-based analysis	**Cost-sensitive weighting was more reliable than SMOTE**; SMOTE reduced deployment validity under extreme imbalance. [6]
Managing class imbalance in the training of a large language model to predict patient selection for total knee arthroplasty	Not surfaced in retrieved evidence	2025	Knee	10.1016/j.knee.2025.02.007	Clinical text / radiology reports	11.8% minority class	Class weighting vs majority-class undersampling vs baseline	Recall, AUROC	**Class weighting outperformed undersampling** for minority-class discrimination and calibration. [7]
Application of machine learning algorithms with balancing techniques for credit card fraud detection: A comparative analysis	Not surfaced in retrieved evidence	2022	Fundamentals and Methods of Machine and Deep Learning: Algorithms, Tools, and Applications	10.1002/9781119821908.ch12	Credit-card fraud data (not health/survey)	Highly imbalanced	Balanced class weight vs SMOTE oversampling	Accuracy, recall, precision, F1, MCC, AUC, AUPRC	**SMOTE performed best overall**, while class weighting achieved the lowest false positives. [8]

Bottom line

Across the retrieved health and survey studies, **cost-sensitive learning/class weights most often wins when the goal is calibrated minority-class discrimination with minimal data distortion**, while **SMOTE tends to help when the minority class is sparse enough that synthetic densification materially improves separation**; several studies also show that **the best choice is dataset-specific rather than universal**. [1] [2] [3] [4] [5] [7] [9]

Confidence level: **High** — the matrix is directly grounded in studies that explicitly compare weighting/cost-sensitive methods against SMOTE or oversampling, though some requested fields (notably authors and exact imbalance ratio) were not surfaced in the retrieved evidence for every paper.

References

1. Gaviria-Chavarro, J.G., Gómez-García, M.Á., Alcaide Leyva, J.M., (...), Zambrano Bermeo, R.N. (2026) Identification of Predictors of Adaptability in Older Adults Based on the Roy Adaptation Model Using Machine Learning. *Journal of Clinical Medicine*. https://doi.org/10.3390/jcm15051709
2. Liu, Y., Li, B., Yang, S., Li, Z. (2024) Handling missing values and imbalanced classes in machine learning to predict consumer preference: Demonstrations and comparisons to prominent methods. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2023.121694

3. de Almeida, F.G., Baggström, A., Andermann, T. (2025) Estimating national Red List statuses for fungi in Sweden - An improved deep learning approach to account for unbalanced data. *Ecological Informatics*. https://doi.org/10.1016/j.ecoinf.2025.103476
4. Laksono, T.A., Andriyani, W. (2025) Imbalanced Data Handling for Stroke Prediction Using Oversampling and Cost-Sensitive Learning. *Proceedings - 2025 8th International Seminar on Research of Information Technology and Intelligent Systems, ISRITI 2025*. https://doi.org/10.1109/ISRITI68345.2025.11393348
5. Traka, E., Papageorgiou, G., Mantzavinis, G., Tjortjis, C. (2026) Comparative Forecasting and Misclassification Analysis Using Health Survey Data. *AI (Switzerland)*. https://doi.org/10.3390/ai7040148
6. Catana, A.-E., Florescu, A. (2026) Resampling strategies for machine learning-based effort overrun risk detection: A controlled factorial study with cost-sensitive evaluation. *International Journal of Optimization and Control: Theories and Applications*. https://doi.org/10.36922/IJOCTA026200080
7. Farrow, L., Anderson, L., Zhong, M. (2025) Managing class imbalance in the training of a large language model to predict patient selection for total knee arthroplasty: Results from the Artificial intelligence to Revolutionise the patient Care pathway in Hip and knEe aRthroplastY (ARCHERY) project. *Knee*. https://doi.org/10.1016/j.knee.2025.02.007
8. Shiksha, S. (2022) Application of machine learning algorithms with balancing techniques for credit card fraud detection: A comparative analysis. *Fundamentals and Methods of Machine and Deep Learning: Algorithms, Tools, and Applications*. https://doi.org/10.1002/9781119821908.ch12
9. Di Martino, F., Delmastro, F., Dolciotti, C. (2023) Explainable AI for malnutrition risk prediction from m-Health and clinical data. *Smart Health*. https://doi.org/10.1016/j.smhl.2023.100429
10. Huang, J. (2025) Initial Diagnosis of Stroke Based on SMOTE- ENN and Hybrid Machine Learning. *Procedia Computer Science*. https://doi.org/10.1016/j.procs.2025.05.115
11. Çınaroğlu, S. (2024) Learning from unbalanced catastrophic out-of-pocket health expenditure dataset: blending SMOTE-boosting with ensemble models. *Journal of Experimental and Theoretical Artificial Intelligence*. https://doi.org/10.1080/0952813X.2022.2143907